Lab 2: Group and create

Today we learned to count per group, to group and calculate, and to create new columns of our own. There are many tools at once, and it takes some time before they feel natural. This lab has two parts, and **you choose**.

Part 1: Practise repeats the steps from today with other columns and other questions. The principle is the same. Do this part if you want to try the tools once more and feel sure about them.

Part 2: Go deeper is for you who felt "I understood this, and I want to go further". The questions are more open, and you combine the tools from today.

I do not expect you to do both parts. It is your own choice, and you are welcome to contact me if you want to talk about which part to take.

You work in your own notebook, with the same file as today: `data/listings.csv`. Call the table `listings`, and write a short comment above each answer, in your own words. If you get stuck: read the error message from the bottom up, print the thing you are working with, and look at the lecture notebook. Then ask.

Part 1: Practise

Repeat the steps from today with other columns. Same principle, new questions.

Tools you need

```
import numpy as np
listings['x'].value_counts()      value_counts(normalize=True)      value_counts().tail(3)
listings.groupby('a')['b'].mean()
listings.groupby('a')['b'].agg(['count', 'mean', 'median'])
listings.groupby('a').agg(n=('id', 'count'), top=('b', 'max'))     # named columns
pd.crosstab(listings['a'], listings['b'], normalize='index')
listings['new'] = listings['x'] > 0                                # True / False column
listings['new'] = np.where(condition, 'yes', 'no')
pd.cut(listings['x'], bins=[0, 1, 6, 29, 500], labels=['a', 'b', 'c', 'd'])
```

Tasks

1 **Count property types**
Load the file. Use `value_counts()` on `property_type` and show only the five most common types. What share of all listings is `Entire rental unit`? (Use `normalize=True`, rounded to three decimals.)

2 **The smallest neighbourhoods**
Use `value_counts()` on `neighbourhood_cleansed`. Which three neighbourhoods have the fewest listings, and how many do they have?

3 **Average beds per room type**
Calculate the average of `beds` for each `room_type`, rounded to two decimals. Then use `agg(['count', 'mean'])` on the same thing. Which average do you trust the least, and why?

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4 **Reviews per neighbourhood**
Group by `neighbourhood_cleansed` and calculate `count`, `mean` and `median` of `number_of_reviews`, rounded to two decimals, sorted by the median. Which neighbourhood has the lowest median and which the highest? For Södermalms the mean is much bigger than the median. What could be the reason?

5 **Your own summary table**
Make a summary per `room_type` with named columns: `n_listings` (count of `id`), `avg_beds` (mean of `beds`) and `max_reviews` (max of `number_of_reviews`). Round to two decimals.

6 **Two categories at once**
Make a crosstab of `neighbourhood_cleansed` and `host_is_superhost` with the share in percent for each neighbourhood (`normalize='index'`, times 100, one decimal). The column `t` means that the host is a superhost. Which neighbourhood has the highest share of superhosts, and which the lowest?

7 **Create new columns**
(a) Create a column `has_reviews` that is `True` when `number_of_reviews` is bigger than 0. How many listings are `False`?
(b) Create a column `popularity` with `np.where`: `'popular'` if the listing has 50 reviews or more, otherwise `'other'`. Count the values.

8 **Turn numbers into groups**
The column `minimum_nights` says how long you have to stay. Make four groups with `pd.cut`: 1 night, 2-6 nights, 7-29 nights and 30 or more nights. Save them as a column `stay` with labels, and count the groups with `value_counts().sort_index()`. How many listings have no group?

On purpose: First run the `cut` with `bins=[1, 6, 29, 500]` and count the missing values with `isna().sum()`. Why are so many missing? Then fix it.

Expected
Task 1: Entire rental unit 2442, Entire home 569, Private room in rental unit 491, Entire condo 339, Entire villa 149. The share is 0.534.
Task 2: Spånga-Tensta 89, Skärholmens 96 and Rinkeby-Tensta 108.
Task 3: Entire home/apt 2.48, Hotel room 1.00, Private room 1.48, Shared room 4.58. The counts are 3680 / 3 / 766 / 24 (listings where the number of beds is known), so the Hotel room average stands on only three listings.
Task 4: the lowest median is Skärholmens (3), the highest is Södermalms (17). Södermalms has mean 59.82.
Task 5: n_listings 3760 / 3 / 787 / 24, avg_beds 2.48 / 1.00 / 1.48 / 4.58, max_reviews 1293 / 79 / 867 / 510.
Task 6: the highest share of superhosts is Södermalms (23.8), the lowest is Bromma (7.1).
Task 7: (a) 644 listings are `False`. (b) other 3805, popular 769.
Task 8: with `bins=[1, 6, 29, 500]` there are 1331 missing (the 1327 listings with 1 night, plus 4 that are missing from the start). After the fix: 1 night 1327, 2-6 nights 2661, 7-29 nights 430, 30+ nights 152, and 4 listings with no group.

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Part 2: Go deeper

Here you combine the tools from today to answer more open questions. There is no new syntax, but you have to think about what the numbers mean.

Tools you need

```
listings.groupby('a').agg(n=('id', 'count'), share=('mask', 'mean'))     # mean of True/False =
pd.qcut(table['x'], 4, labels=['few', 'some', 'many', 'most'])
pd.qcut(table['x'], 4).cat.categories              # the borders of the groups
listings.groupby(['a', 'b']).agg(n=('id', 'count'), avg=('x', 'mean'))
table[table['n'] >= 30]                            # filter an aggregated table
```

Tasks

1 **Who has never been reviewed?**
Use the column `has_reviews` from Part 1 (or create it again). What share of all listings has no reviews? Then calculate the **share without reviews** in each neighbourhood. Which neighbourhoods have the lowest and the highest share? (Hint: the mean of `True`/`False` is a share, so think about which of the two you want the mean of.)

2 **Quartiles of reviews**
Keep only the listings with at least one review, and call the result `reviewed`. Divide `number_of_reviews` into four groups with `pd.qcut`, with the labels `few`, `some`, `many` and `most`. Count the groups. Are they exactly the same size? Why or why not? Then print the **borders** of the four groups (`.cat.categories`).

3 **Do more reviews mean a better rating?**
Group `reviewed` by your four review groups and calculate the average `review_scores_rating`. Does the rating go up when there are more reviews? Write in a comment what you **can** and **cannot** conclude from the table.

4 **The biggest groups, but only if they are real**
Group by `neighbourhood_cleansed` and `room_type` at the same time. Calculate the number of listings and the average `accommodates`. Sort by the average, biggest first, and look at the top three. Then keep only the combinations with **at least 30 listings** and look at the top three again. Why is the second list more trustworthy?

5 **Your own question**
Come up with **one question** that you can answer by grouping, counting or creating a new column. Write the question as a comment, answer it with code, and finish with a comment about what the answer does **not** tell you. That last part is the most important one.



Expected
Task 1: 644 listings have no reviews, which is 14.1 percent. The lowest share is Södermalms (0.076) and the highest is Hässelby-Vällingby (0.270).
Task 2: the groups have 1010, 1032, 914 and 974 listings. They are not equal, because many listings have exactly the same number of reviews and ties cannot be split. The borders are 1, 4, 13, 36 and 1293.
Task 3: the averages are about 4.79, 4.84, 4.83 and 4.80. There is no clear pattern.
Task 4: there are 35 combinations. Without the filter the top is Östermalms / Shared room (9.00 guests, but only 4 listings). With at least 30 listings 27 combinations are left, and the top is Spånga-Tensta / Entire home/apt (5.33 guests, 54 listings).

Stuck? Read the error message from the bottom up, print the data you are working with, and then ask.